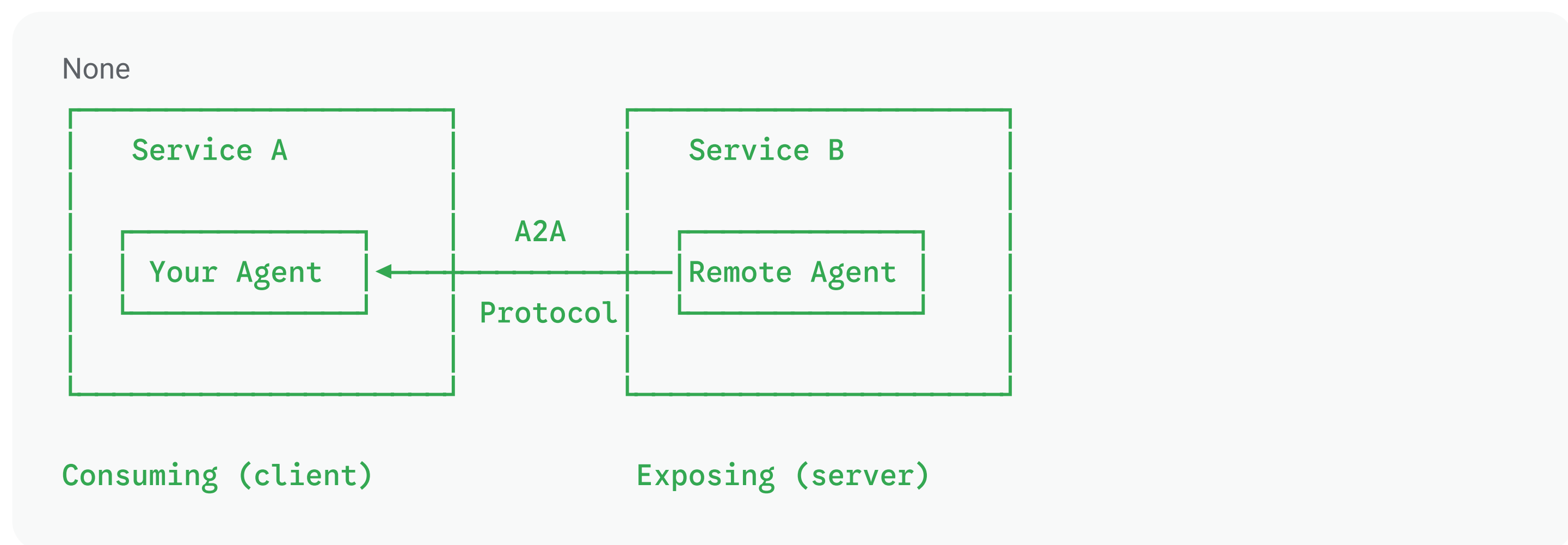


How **A2A** works (conceptual)

Two sides of A2A



Exposing: Making an agent available as a network service

Consuming: Calling a remote agent from your application

The agent card

Every A2A agent has an **agent card**—a JSON file describing its capabilities:

JSON

```
{
  "name": "research_specialist",
  "description": "Conducts research on scientific topics",
  "capabilities": ["research", "summarization"],
  "url": "https://research.example.com"
}
```



The agent card is how consuming agents know what the remote agent can do.

A2A in ADK (preview)

ADK provides built-in A2A support:

```
Python
# Exposing (server side)
from google.adk.a2a import A2AServer

server = A2AServer(agent=my_agent, port=8001)
server.start()

# Consuming (client side)
from google.adk.agents import RemoteA2aAgent

remote = RemoteA2aAgent(
    name='remote_research',
    agent_card_source='http://research-service:8001/.well-known/agent-card.json'
)
```

Note: This is just a preview. Full implementation is covered in the [Advanced Learning Path](#).

Resources for learning more

When you're ready to implement distributed agents:

Official documentation

- A2A Protocol: <https://a2a-protocol.org/>
- ADK A2A Guide: <https://google.github.io/adk-docs/a2a>

ADK A2A tutorials

- Exposing agents: [ADK Quickstart - Exposing](#)
- Consuming agents: [ADK Quickstart - Consuming](#)

Key takeaways

A2A Protocol opens up powerful enterprise-scale possibilities for agent coordination, but it’s not needed for most learning and development scenarios. Focus on mastering local agent orchestration first—when you encounter cross-service or cross-team agent coordination needs, A2A will be waiting.

When to use A2A

Use A2A When	Use local agents when
Microservices architecture	Single application
Cross-team/organization	Same team
Different programming languages	Same codebase
Independent scaling needed	Performance-critical

Learning path

Your Stage	Recommendation
Learning ADK	Focus on local agents (courses 1-9)
Building production apps	Consider A2A if you have distributed needs
Enterprise scale	A2A becomes valuable

